

Lesson 7: Newton's Second Law with Normal-Tangential coordinates

Reminder, Newton's second law is

$$\sum \vec{F} = m\vec{a}$$

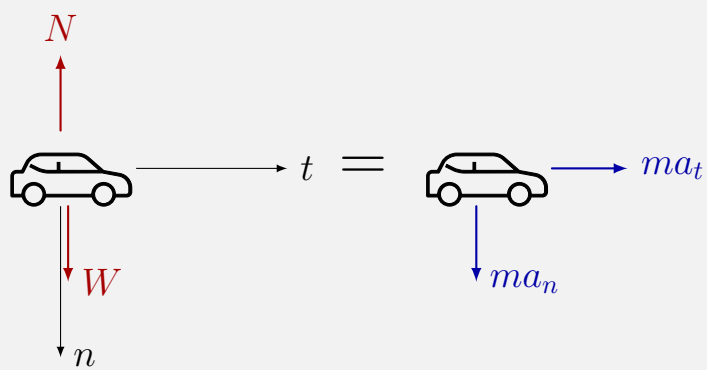
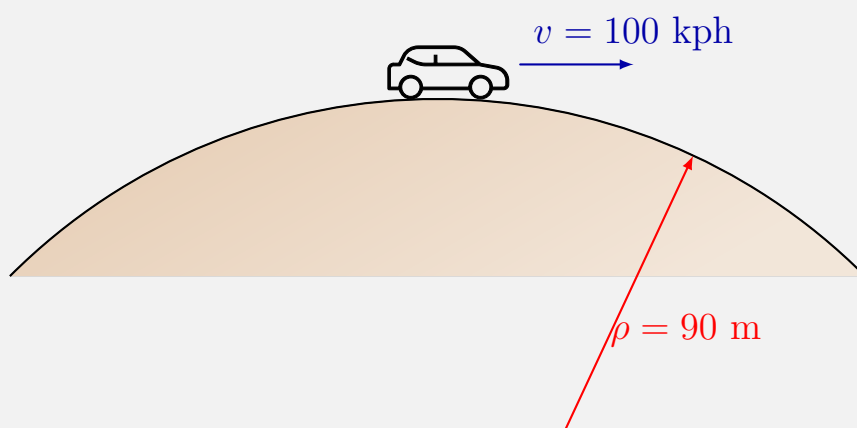
In normal-tangential coordinates this becomes

$$\sum F_t = m \frac{dv}{dt} \quad \sum F_n = m \frac{v^2}{\rho}$$

Sample Problem 7.1

A 1000 kg car travels over a hill with a constant speed of 100 kilometers per hour. The top of the hill can be approximated as a circle with a 90 m radius. What is the normal force the road exerts on the car as it crests the hill? How fast would the car have to be going to get airborne?

7.1 – cont



7.1 – cont

Newton's second law in the normal direction is

$$\begin{aligned}\sum F_n &= m \frac{v^2}{\rho} \\ W - N &= m \frac{v^2}{\rho} \\ N &= W - m \frac{v^2}{\rho}\end{aligned}$$

The weight of the vehicle is

$$W = 1000 \text{ kg} (9.81 \text{ m/s}^2) = 9810 \text{ N}$$

The normal component of the acceleration is

$$a_n = \frac{v^2}{\rho} = \frac{(27.78 \text{ m/s})^2}{90 \text{ m}} = 8.575 \text{ m/s}^2$$

$$N = W - m \frac{v^2}{\rho}$$

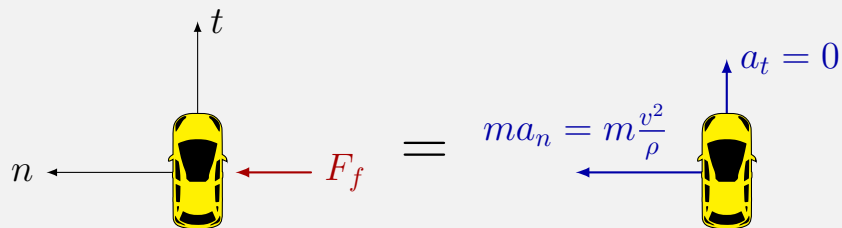
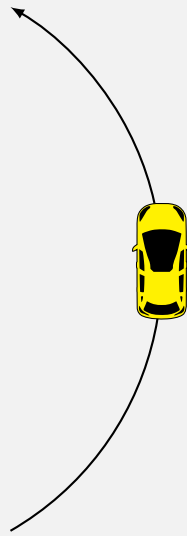
$$N = 9810 \text{ N} - 1000 \text{ kg}(8.575 \text{ m/s}^2) = 1235 \text{ N}$$

To find the speed when the car becomes airborne we set $N = 0$.

$$\begin{aligned}W &= m \frac{v^2}{\rho} \implies v = \sqrt{g\rho} \\ v &= \sqrt{9.81 \text{ m/s}^2(90 \text{ m})} = 29.71 \text{ m/s} = 107 \text{ kph}\end{aligned}$$

Sample Problem 7.2

A 2500 pound car is traveling 40 feet per second. The car's tires have a coefficient of friction with the road of $\mu_s = 0.9$. If the car is maintaining a constant speed, what is the minimum radius of curvature before slipping? Assuming the car is speeding up at a rate of 10 ft/s^2 , what is the minimum radius of curvature before slipping?



7.2 – cont

Newton's second law in the normal direction is

$$\sum F_n = m \frac{v^2}{\rho}$$

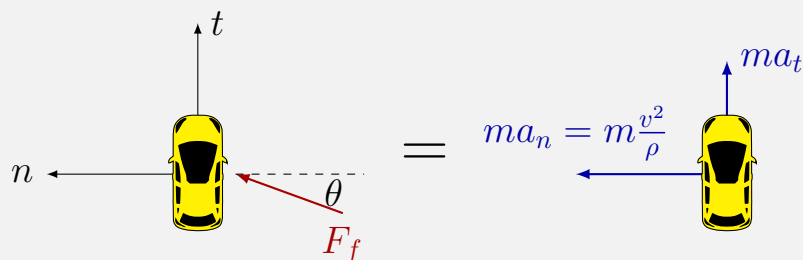
$$F_f = m \frac{v^2}{\rho}$$

The friction force is $F_f = \mu_s N$ where the normal force is not shown in the above diagram. If it was it would be pointing out of the page. In this problem $N = W = 2500$ lb. The friction force is therefore $F_f = (0.9)2500 = 2250$ lb. The mass of the car is $m = W/g = 2500/32.2 = 77.64$ slug.

$$F_f = m \frac{v^2}{\rho}$$

$$2250 = 77.64 \frac{40^2}{\rho} \Rightarrow \boxed{\rho = 55.2 \text{ ft}}$$

If the car also has a tangential acceleration the force of friction is no longer acts solely in the normal direction. We have drawn the friction force vector at an unknown angle θ . The normal component of the friction force is $F_f \cos \theta$ and the tangential component is $F_f \sin \theta$.



7.2 – cont

Newton's second law in the tangential directions is

$$\begin{aligned}\sum F_t &= m \frac{dv}{dt} \\ F_f \sin \theta &= 77.64 \text{ slug} (10 \text{ ft/s}^2) \\ 2250 \text{ lb} \sin \theta &= 776.4 \text{ lb} \\ \sin \theta &= \frac{776.4}{2250} \implies \theta = 20.2^\circ\end{aligned}$$

Newton's second law in the normal directions is

$$\begin{aligned}\sum F_n &= m \frac{v^2}{\rho} \\ F_f \cos \theta &= 77.64 \frac{40^2}{\rho} \\ 2250 \cos 20.2^\circ &= 77.64 \frac{40^2}{\rho} \implies \rho = 58.83 \text{ ft}\end{aligned}$$

A quicker way to arrive at the answer without the intermediate step of calculating the angle is by recognizing that the force of friction and total acceleration vector must point in the same direction. We can therefore set the magnitude of the net force equal to mass times the magnitude of the net acceleration.

7.2 – cont

$$F_f = m\sqrt{a_n^2 + a_t^2}$$

$$\mu mg = m\sqrt{\left(\frac{v^2}{\rho}\right)^2 + (a_t)^2}$$

$$\mu g = \sqrt{\left(\frac{v^2}{\rho}\right)^2 + (a_t)^2}$$

$$(\mu g)^2 = \left(\frac{v^2}{\rho}\right)^2 + (a_t)^2$$

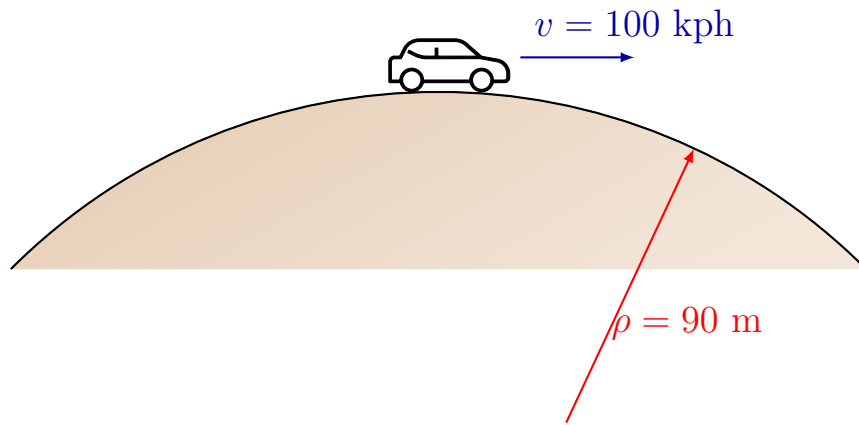
$$(\mu g)^2 - a_t^2 = \left(\frac{v^2}{\rho}\right)^2$$

$$\rho^2 = \frac{v^4}{(\mu g)^2 - a_t^2}$$

$$\rho = \sqrt{\frac{v^4}{(\mu g)^2 - a_t^2}} = \sqrt{\frac{40^4}{(0.9 \times 32.2)^2 - 10^2}} = 58.83 \text{ ft}$$

Sample Problems

1. A 1000 kg car travels over a hill with a constant speed of 100 kilometers per hour. The top of the hill can be approximated as a circle with a 90 m radius. What is the normal force the road exerts on the car as it crests the hill? How fast would the car have to be going to get airborne?



2. A 2500 pound car is traveling 40 feet per second. The car's tires have a coefficient of friction with the road of $\mu_s = 0.9$. If the car is maintaining a constant speed, what is the minimum radius of curvature before slipping? Assuming the car is speeding up at a rate of 10 ft/s^2 , what is the minimum radius of curvature before slipping?

